RTS23 – PHYTONS IN VG: VIBRATIONAL MODES OF COUPLINGS IN EXTREME REGIMES

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ABSTRACT

Phytons are not fundamental particles, but **vibrational modes of the couplings** K_{ij} that appear only in extreme regimes – when the tension on the links reaches critical values and the Network behaves locally as a discrete quantum system. The episode explains why, in normal regimes, the tension on couplings is a **global resultant**, continuous and unpredictable, and phytons are merely a **local approximation** useful in black holes, white holes, phase transitions, and critical nodal density points. Phytons are thus **quanta of effective tension**, not fundamental entities of the Network.

Keywords: phyton, nodal coupling, effective tension, extreme regime, local quantization, black holes, white holes, RTS.

1. INTRODUCTION – WHAT IS A PHYTON?

In RTS, a phyton is not an independent particle, as in some speculative theories. It is a **vibrational mode** of the coupling K_{ij} – that is, a **localized excitation** of the link between two nodes, which appears only when the **tension on that link reaches critical values**. In normal regimes, tension is a global resultant, continuous and unpredictable, and phytons have no physical meaning.

2. TENSION ON COUPLINGS – A GLOBAL PROPERTY

In RTS, the tension on a coupling K_{ij} is not a local property. It is a **resultant of interactions with all other nodes** – an unimaginably complex superposition of vibrations that influence each other. This tension is, in general, **continuous and unpredictable**, and any attempt to describe it through discrete modes would be a crude approximation.

What does this mean for RTS?

· There are no "fundamental quanta" of tension on couplings.

· Tension is an emergent property that depends on the entire Network.

· Quantization is possible only in **limit regimes**, when the Network stabilizes locally.

3. WHEN PHYTONS APPEAR – EXTREME REGIMES

Phytons appear only when:

· **Nodal density reaches critical values** (such as near a black hole or a white hole).

· Couplings become strong enough to generate **discrete vibrational modes** on the links.

· **The resultant tension stabilizes** sufficiently to allow a local quantum description.

In these conditions, the link behaves like a **quantum string** with well-defined frequencies, and phytons are the **quanta of these frequencies**.

What phytons are not:

· They are not fundamental particles.

· They are not universal carriers of the torsion field.

· They are not necessary to describe the Network in normal regimes.

What phytons are:

· **Localized vibrational modes** of couplings in extreme regimes.

· **Computational tools** to describe the Network's behavior near singularities.

4. PHYTONS AS A LOCAL APPROXIMATION

In extreme regimes, the Network behaves locally as a discrete quantum system. In this approximation:

· The effective tension on a link can be written as a sum of discrete modes.

· Each mode corresponds to a **phyton**.

· Phytons are not real in a fundamental sense – they are **tools** to describe a limit regime.

Analogy with a guitar string:

· A guitar string has a continuous spectrum of possible tensions.

· When plucked, it produces **discrete modes** (notes).

· These modes are not "particles" – they are properties of the string in that regime.

Likewise, phytons are the **notes that couplings play** when they are stretched to the limit.

5. WHAT'S NEW IN SCIENCE?

RTS23 introduces a new understanding of phytons as **localized vibrational modes of couplings**, not as fundamental particles. The episode clarifies:

· The difference between normal regimes (continuous, global tension) and extreme regimes (locally quantized tension).

· Phytons as **theoretical tools** for describing black holes, white holes, and critical points.

· Phytons are not fundamental entities, but **local approximations** valid only under extreme conditions.

6. CONCLUSION – PHYTONS AS TOOLS, NOT AS ENTITIES

Phytons are not the discovery of a new particle. They are a **theoretical tool** useful for describing **couplings in extreme regimes** – whether absorption (BH), emission (WH), or any other point where the Network reaches a critical nodal density threshold. In normal regimes, the tension on couplings remains a global, continuous, and unpredictable resultant, and phytons have no physical meaning.

What does this mean for RTS?

· Phytons are a **local approximation**, not a fundamental property.

· They appear only in extreme absorption or emission regimes, at critical points.

· In the rest of the Network, tension is **too complex** to be reduced to quanta.